

AGENTIC AI · AGENTIC RAG · REFERENCE BUILD

footnote¹

An AI research agent for your documents. It decides what to search for, searches again when results are thin, and every claim links to the exact sentence it came from.

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The dangerous answer is the one that reads well.

"Chat with our documents" is the first thing most companies ask an LLM to do.

The usual build is a fixed pipeline: search once with the user's words, paste the results into a prompt. A two-part question gets half an answer.

And it fails quietly. The answer is fluent, and one sentence in it came from nowhere. Someone acts on that sentence.

One rule: an answer you cannot check is not an answer.

- An agent plans the research: it picks queries, splits the question, retries.
- It may use only what its tools returned.
- Every claim carries a citation the API has validated.
- One click shows the source sentence, highlighted in place.
- When the documents do not cover it, it says so.

An answer, with its receipts.

footnote¹232 documents · 1432 passages · demo (scripted replies)

GitHub

Demo mode. No API key is set, so the model's replies are scripted for the example questions. The searches, the passages and the citation matching are real.

How much annual leave do I earn, and can I carry it over?

Ask

Search tool uses:

hybrid

 keyword only vector only

How much annual leave do I earn, and can I carry it over?

How do I get reimbursed for a conference?

What is the dress code for the office Christmas party?

Annual leave builds up with your length of federal service¹: fewer than 3 years earns 4 hours per pay period, 3 to 15 years earns 6, and 15 or more earns 8.¹ The rate goes up on the first full pay period after your work anniversary.¹

You can carry over at most 240 hours of annual leave from one leave year to the next.² Anything above that shows as "Use or Lose": hours you forfeit at the end of the leave year if you do not take them. It applies to annual leave only.²

model turns 2searches 2total 3.0 s passages cited 2 of 9answer parts with a citation 5 of 5

WHAT THE AGENT DID

Two parts here: how leave builds up, and the carry-over limit. Searching each.

1. search "annual leave accrual rate hours per pay period"

5 new passages · 302 ms

2. search "annual leave carry over maximum use or lose"

4 new passages · 6 ms

PASSAGES IT WAS SHOWN

1Leave types › Annual leave

search 1 · keyword #1 · vector #1 · source

You may use annual leave for any reason. Annual leave accrues based on how long you've worked for the federal government:

- Fewer than 3 years: 4 hours per pay period

- Between 3 and 15 years: 6 hours per pay period

- 15+ years: 8 hours per pay period

Your leave increases on the first full pay period following your work anniversary. You can refer to either HR Links or the bottom of your paystub for your leave balances.

You can carry over a maximum of 240 annual leave hours, from one leave year to the next. If you've transferred to TTS from another federal agency, your accrued annual leave will transfer with you.

2Leave types › Use or Lose

search 2 · keyword #1 · vector #1 · source

Use or Lose is the number of annual leave hours you would forfeit at the end of the leave year, if you do not use any of your annual leave hours between now and then. As noted above, there is a 240 hour limit on the number of hours you can carry over from one leave year to the next. Use or Lose only applies to annual leave.

For example: You look in HRLinks and see "182 Hours (96 Hours Use or Lose)." Today, January 25th, 2022, you have 182 hours of annual leave available. If you continued to accrue annual leave

A two-part question. The agent ran two searches, in the handbook's wording, one per part.

Right: its working note, each query, and the passages that came back. Cited ones rise to the top; the rest stay visible so you can see what it ignored.

Left: the answer, streamed. Each chip is a citation returned by the API.

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Click a claim. See the sentence.

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Search results go back to the model one sentence per block. The API returns which result and which sentences each claim rests on, and checks them. The model cannot cite text that is not there.

Prompting a model to type "[1]" gives markers that look right and are sometimes wrong. This does not.

"That is not in the documents."

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What is the dress code for the office Christmas party?

Ask

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How much annual leave do I earn, and can I carry it over?

How do I get reimbursed for a conference?

What is the dress code for the office Christmas party?

The handbook does not cover a Christmas party, or a dress code for any office event, so I can't answer that from it. The closest it gets is everyday office wear. In Chicago there is no official dress code and business casual is what you will mostly see. ¹ In New York people range from casual t-shirts to business casual, and suits are rare. ² For a specific event, ask whoever is organising it.

model turns **3** searches **2** total **2.9 s** passages cited **2 of 9** answer parts with a citation **2 of 4**

WHAT THE AGENT DID

Looking for a dress code first.

1. search "office dress code"

5 new passages · 9 ms

Nothing about events. Trying the party angle before giving up.

2. search "holiday party office celebration"

4 new passages · 7 ms

PASSAGES IT WAS SHOWN

2 New York City · How do people dress?

search 1 · keyword #2 · vector #1 · [source](#)

TTS folk in the office usually range from casual t-shirts to business casual (jeans and a collared shirts, or blouse). Suits would not be out of place, but are not universal (and pretty rare for TTS folks when we come into the office). The office can be a little cold so cardigans, blazers, or sportcoats (sweaters and hoodies are pretty common, too). On your first day, folks from other parts of GSA will mostly be dressed business casual. Your first step onboarding includes being sworn in as a civil servant, which in the New York office likely takes place while looking over New York Harbor with a view of the Statue of Liberty. Some felt it was hokey, others were moved, many felt both. Several were glad they dressed for the occasion. We care more that you are comfortable and confident as yourself rather than following someone's particular rules of formality. And yes, you should totally feel free to dye your hair in rainbow colors and rock that undercut. You be you.

1 Chicago · Who can I contact if I have questions about the office space?

search 1 · keyword #1 · vector #2 · [source](#)

There is a health center (Federal Occupational Health) near the southeast corner of the 3rd floor.

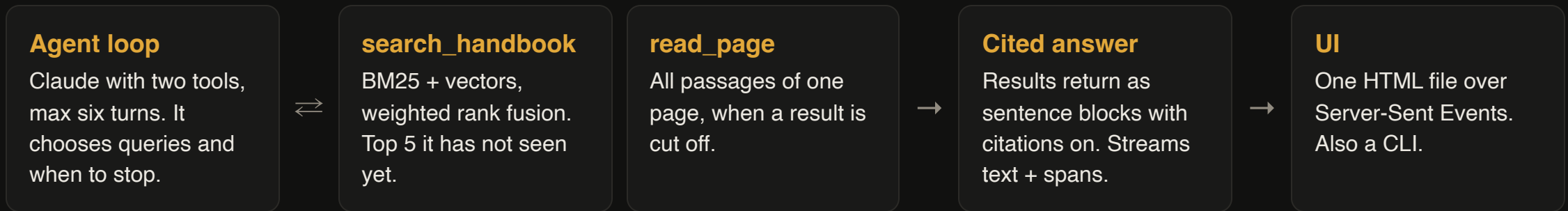
A more comprehensive guide of Chicago office amenities and policies can be found in a slideshow on this Insite page.

The handbook has nothing on Christmas parties. The agent searched for a dress code, tried the party angle, then stopped. The answer says what is missing and cites the nearby material it did find.

The prompt explains why guessing is harmful rather than shouting "never". The eval set includes ten questions like this one.

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An agent loop over a search tool. No framework.



- **No vector database.** 2 MB of vectors; a numpy multiply takes under a millisecond. Worth revisiting past a few hundred thousand passages.
- **Local embeddings.** No second API key, no per-query cost, documents stay on the machine.
- **Hybrid search.** Vectors miss exact terms; keywords miss paraphrase. Rank fusion needs no score calibration.
- **Headings travel with the passage.** "You can carry over 240 hours" means nothing to search without "Annual leave".

The eval caught my first hybrid search losing to plain vectors.

SEARCH MODE	CORRECT PASSAGE IS #1	IN TOP 3	IN TOP 8	MRR
Keyword only (BM25)	28.8%	55.9%	64.4%	0.415
Vector only	44.1%	67.8%	84.7%	0.581
Hybrid, equal weights (first attempt)	45.8%	71.2%	74.6%	0.580
Hybrid, 70/30 towards vectors	45.8%	74.6%	84.7%	0.608

One search call per question, the floor the agent starts from. 59 questions written by Claude from sampled passages, told to avoid the passage's wording. Small, synthetic, and the weights were tuned on it, so read the direction, not the decimals. The point: without the eval, the equal-weight version would have shipped.

What is measured, and what is not yet.

Measured: the search tool, on 59 questions (previous slide), and the pipeline code, by unit tests in CI.

Built, not yet run: the agent eval. 59 answerable questions plus 10 the handbook cannot answer. A second model grades each answer as correct, partial, wrong or abstained.

No answer-quality numbers are claimed until that suite has run on the agentic version.

- Was the right passage ever retrieved, across all searches?
- Did the answer cite it?
- What share of the answer text carries a citation?
- Did it abstain on the unanswerable ones?
- Searches per question, latency, cost.

Screenshots in this deck are from demo mode: scripted model turns, real searches, real citation matching.

WHAT THIS SHOWS

Skills on display

- Agentic AI: a tool-using research loop with step limits, retries and a stop rule
- RAG end to end: cleaning, chunking, embedding, hybrid search, rank fusion
- Claude API: tool use, citations on tool results, streaming, structured outputs
- Evaluation: synthetic question sets, search metrics, LLM-as-judge, abstention tests
- Product sense: the feature is trust, so the UI is built around checking
- Engineering restraint: SQLite + numpy instead of infrastructure nobody needs yet
- Python, FastAPI, Server-Sent Events, pytest, GitHub Actions

Next: cross-encoder reranking, follow-up question rewriting, a human-written eval set, passage-level access control.

THANK YOU

Ibrar Yunus

AI engineer. I build LLM systems that can be checked.

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